Theorem 3: Let A and B denote matrices whose sizes are appropriate for the following sums and products.

a) $(A^T)^T = A$ b) $(A+B)^T = A^T + B^T$ c) For any scalar r, $(rA)^T = rA^T$

d) $(AB)^T = B^T A^T$.

Proof: Let A be a $m \times n$ matrix. $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$

a) $A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \cdots & a_{m2} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{m3} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & a_{3n} & & a_{mn} \end{bmatrix}$ (From defn. of transpose)

$(A^T)^T = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & & a_{mn} \end{bmatrix} = A$ (From defn. of transpose)

$\therefore \boxed{(A^T)^T = A}$ $\square$

b) Let B be a $m \times n$ matrix. $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & \cdots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \cdots & b_{2n} \\ b_{31} & b_{32} & b_{33} & \cdots & b_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & b_{m3} & \cdots & b_{mn} \end{bmatrix}$

$(A+B) = \begin{bmatrix} (a_{11}+b_{11}) & (a_{12}+b_{12}) & (a_{13}+b_{13}) & \cdots & (a_{1n}+b_{1n}) \\ (a_{21}+b_{21}) & (a_{22}+b_{22}) & (a_{23}+b_{23}) & \cdots & (a_{2n}+b_{2n}) \\ (a_{31}+b_{31}) & (a_{32}+b_{32}) & (a_{33}+b_{33}) & \cdots & (a_{3n}+b_{3n}) \\ \vdots & & & & \\ (a_{m1}+b_{m1}) & (a_{m2}+b_{m2}) & (a_{m3}+b_{m3}) & \cdots & (a_{mn}+b_{mn}) \end{bmatrix}$

(By defn. of matrix addition)